Smart Learning Platform - Complete Algorithm Documentation

Executive Summary

The **Smart E-Learning Platform** implements a sophisticated **hybrid personalization system** that combines:

• **Difficulty-based level gating** (strict progression model)

• **Interest-based course filtering** (recommendation engine)

• **Adaptive remediation** (personalized learning paths based on performance)

• **Gamification rewards** (motivation through stars and badges)

• **Category-based sequencing** (structured curriculum progression)

This document presents the complete algorithmic framework powering student personalization, course progression, and adaptive learning paths.

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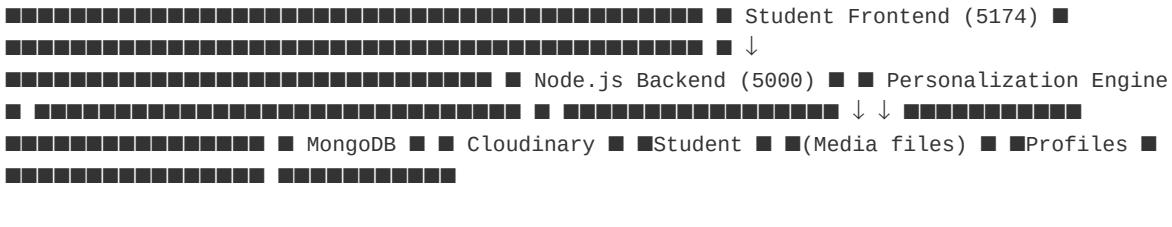
System Overview

Architecture



Key Data Models

```
{ "userId": "...", "learningStyle": "visual|text|literacy|numeracy", "interests": ["tag1",
"tag2"], "difficultyPreference": "beginner|intermediate|advanced", "signLanguage":
"Tanzanian Sign Language", "stars": 0, "badges": [], "onboardingComplete": false,
"levelStatus": { "beginner": false, "intermediate": false, "advanced": false } }
```

```
{ "studentId": "...", "courseId": "...", "completedVideos": [], "totalTimeSpent": 0,
"completionStatus": "not_started|in_progress|completed", "passedLevelQuiz": [],
"quizAttempts": [], "needsRemedial": false, "remedialContent": [], "isCourseCompleted":
false }
```

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Personalization Attributes

Collected During Onboarding

Attribute	Type	Values	Purpose
Learning Style	Enum	visual, text, literacy, numeracy	Content format preference
Interests	Array	Tags/categories	Course recommendation filtering
Difficulty Preference	Enum	beginner, intermediate, advanced	Course level access control
Sign Language	String	Tanzanian Sign Language (default)	Accessibility/localization
Avatar	Number	1-N	User profile customization

Derived During Learning

Attribute	Type	Purpose
Stars	Number	Gamification (10 per quiz pass)
Badges	Array	Achievement tracking (4 types)
Level Status	Object	Boolean flags per difficulty level
Course Progress	Ref	Video completion, quiz attempts, remedial paths

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Core Algorithms

Algorithm 1: Personalization Data Collection & Storage

ALGORITHM CollectPersonalization(studentId, surveyData): INPUT: - studentId: unique student identifier - surveyData: {learningStyle, interests, difficultyPreference, signLanguage} PROCESS: 1. Load student document from MongoDB 2. Update fields: - learningStyle ← surveyData.learningStyle - interests ← surveyData.interests - difficultyPreference ← surveyData.difficultyPreference - signLanguage ← surveyData.signLanguage - onboardingComplete ← true 3. Set timestamps (createdAt, updatedAt) 4. Save to MongoDB OUTPUT: - Updated student profile - Confirmation response TIME COMPLEXITY: O(1) - Single document write SPACE COMPLEXITY: O(1) - Fixed document structure

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Algorithm 2: Course Filtering by Personalization

ALGORITHM FilterCoursesByPersonalization(studentId): INPUT: - studentId: unique student identifier PROCESS: 1. Load student profile from MongoDB - Get: difficultyPreference, interests 2. PRIMARY FILTER - Difficulty Level: courses ← DB.Course.find({ level: student.difficultyPreference ← EXACT MATCH }) 3. SECONDARY FILTER - Interests (Optional): IF student.interests is NOT EMPTY: courses ← courses.filter(course => course.tags ∩ student.interests ≠ ∅) 4. TERTIARY FILTER - Category Progression: courses ← courses.sort({ category: ASC, order: ASC, level: ASC }) 5. Return filtered courses OUTPUT: - Array of courses matching criteria TIME COMPLEXITY: O(n) - Database query with filters SPACE COMPLEXITY: O(m) - m courses matching filters

Student: {difficultyPreference: "beginner", interests: ["math", "numeracy"]} Courses Available: ✓ Beginner Math 101 (level: beginner, tags: [math, numeracy]) ✓ Beginner Numeracy Basics (level: beginner, tags: [numeracy]) ✗ Intermediate Algebra (level: intermediate) ← Different level ✗ Beginner English (level: beginner, tags: [english]) ← No interest match

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Algorithm 3: Level Access Control (Gating)

ALGORITHM CheckLevelAccess(studentId, courseId): INPUT: - studentId: unique student identifier - courseId: course to access PROCESS: 1. Load course from MongoDB - Get: course.level 2. IF course.level == "beginner": RETURN ACCESS_GRANTED ✓ 3. ELSE IF course.level == "intermediate": previousLevel ← "beginner" prevLevelQuizPassed ← CheckQuizPassed(studentId, previousLevel) IF NOT prevLevelQuizPassed: RETURN ACCESS_DENIED (403 Forbidden) Message: "You must pass Beginner quiz first" ELSE: RETURN ACCESS_GRANTED ✓ 4. ELSE IF course.level == "advanced": previousLevel ← "intermediate" prevLevelQuizPassed ← CheckQuizPassed(studentId, previousLevel) IF NOT prevLevelQuizPassed: RETURN ACCESS_DENIED (403 Forbidden) Message: "You must pass Intermediate quiz first" ELSE: RETURN ACCESS_GRANTED ✓ OUTPUT: - Access permission (boolean) - Status message (string) TIME COMPLEXITY: O(1) - Level enum comparison SPACE COMPLEXITY: O(1) - Fixed level hierarchy

FUNCTION CheckQuizPassed(studentId, level): progress ← DB.CourseProgress.findOne({studentId}) passedQuizzes ← progress.passedLevelQuiz.filter(q => q.level == level) RETURN (passedQuizzes.length > 0 AND passedQuizzes[0].percentage >= 80)

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Algorithm 4: Video Progress Tracking

ALGORITHM UpdateVideoProgress(studentId, courseId, subsectionId, videoWatchTime): INPUT: - studentId: student identifier - courseId: course identifier - subsectionId: video/subsection identifier - videoWatchTime: seconds watched PROCESS: 1. Load course content subsection ← DB.SubSection.findById(subsectionId) videoDuration ← subsection.duration 2. Calculate minimum watch time requirement MIN_WATCH_TIME ← videoDuration × 0.8 (80% of duration) 3. Load or create courseProgress progress ← DB.CourseProgress.findOrCreate({ studentId, courseId }) 4. Check if watch time meets

threshold IF videoWatchTime >= MIN_WATCH_TIME: progress.completedVideos.push(subsectionId) progress.totalTimeSpent += videoWatchTime 5. Calculate course completion status totalVideos ← DB.SubSection.count({courseId}) completedCount ← progress.completedVideos.length percentCompletion ← (completedCount / totalVideos) × 100 IF percentCompletion == 0: progress.completionStatus ← "not_started" ELSE IF percentCompletion == 100: progress.completionStatus ← "completed" ELSE: progress.completionStatus ← "in_progress" 6. Save progress to MongoDB OUTPUT: - Updated courseProgress document - Current completion percentage - Status update TIME COMPLEXITY: O(1) - Direct document operations SPACE COMPLEXITY: O(v) - v = number of completed videos

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Algorithm 5: Quiz Submission & Adaptive Progression

ALGORITHM SubmitQuiz(studentId, quizId, answers): INPUT: - studentId: student identifier - quizId: quiz identifier - answers: student's selected answers PROCESS: 1. Load quiz and course information quiz ← DB.Quiz.findById(quizId) course ← DB.Course.findById(quiz.courseId) level ← course.level category ← course.category 2. Grade the quiz correctCount ← 0 FOR EACH question IN quiz.questions: IF answers[question.id] == question.correctAnswer: correctCount++ totalQuestions ← quiz.questions.length percentage ← (correctCount / totalQuestions) × 100 3. Load or create courseProgress progress ← DB.CourseProgress.findOrCreate({ studentId, courseId: quiz.courseId }) 4. Record quiz attempt progress.quizAttempts.push({ quizId, percentage, attemptedAt: NOW() }) ■■ ■ DECISION POINT: PASS/FAIL ■ ■■ 5A. IF percentage >= 80%: ■ QUIZ PASSED ■■■■■■■■■■■■■■■■■■■■■■■■■■ a) Mark course completion progress.isCourseCompleted ← true progress.completionStatus ← "completed" b) Record passing quiz progress.passedLevelQuiz.push({ quizId, level, percentage, passedAt: NOW() }) c) Award gamification student.stars += 10 d) Find next course in progression nextCourse ← FindNextCourse(studentId, category, level) IF nextCourse EXISTS: DB.CourseProgress.create({ studentId, courseId: nextCourse.id, completionStatus: "not_started" }) AUTO_ENROLL(studentId, nextCourse.id) e) Check if level complete levelCourses ← DB.Course.find({ category, level }) completedCourses ← Count courses where progress.passedLevelQuiz includes them IF completedCourses == levelCourses.length: student.levelStatus[level] ← true AwardBadge(studentId, "level_completion", level) f) Check for perfect score badge IF percentage == 100: AwardBadge(studentId, "perfect_score") g) Save updates student.save() progress.save() RETURN { status: "PASSED", percentage, nextCourse, starsEarned: 10, message: "Congratulations! Moving to next course." } 5B. ELSE IF percentage < 80%: ■ QUIZ FAILED ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ a) Activate remedial mode progress.needsRemedial ← true b) Load remedial content remedialSubsections ← DB.SubSection.find({ courseId: quiz.courseId, isRemedial: true }) c) Assign remedial path progress.remedialContent ← remedialSubsections.map(s => s.id) d) Lock next course nextProgress ← DB.CourseProgress.findOne({ studentId, courseId: nextCourse.id }) IF nextProgress EXISTS: nextProgress.locked ← true nextProgress.save() e) Save progress progress.save() RETURN { status: "FAILED", percentage, message: "Please complete remedial content and retry", remedialCount: remedialSubsections.length } OUTPUT: - Quiz result (pass/fail) - Next course assignment (if pass) - Remedial content assignment (if fail) - Updated student gamification TIME COMPLEXITY: O(q + c) - q questions, c courses in category SPACE COMPLEXITY: O(r) - r remedial subsections

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Algorithm 6: Remedial Content Path

ALGORITHM GetNextRemediableContent(studentId, courseId): INPUT: - studentId: student identifier - courseId: course identifier PROCESS: 1. Load courseProgress progress ← DB.CourseProgress.findOne({ studentId, courseId }) 2. Check remedial status IF NOT progress.needsRemedial: RETURN NULL (No remedial needed) 3. Find next uncompleted remedial subsection FOR EACH subsectionId IN progress.remedialContent: subsection ← DB.SubSection.findById(subsectionId) IF subsectionId NOT IN progress.completedVideos: RETURN subsection (Return first uncompleted remedial) 4. All remedial content completed IF

all remedial subsections completed: progress.needsRemedial ← false progress.remedialContent ← [] progress.save() RETURN { status: "REMEDIAL_COMPLETE", message: "Ready to retake quiz", quizId: course.quizId } OUTPUT: - Next remedial subsection OR - Completion confirmation TIME COMPLEXITY: O(r) - r remedial subsections SPACE COMPLEXITY: O(1)

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Algorithm 7: Next Course Recommendation

ALGORITHM FindNextCourse(studentId, category, level): INPUT: - studentId: student identifier - category: current course category - level: current course level PROCESS: 1. Load student progress history allProgress ← DB.CourseProgress.find({studentId}) 2. Get all completed courses completedCourses ← allProgress.filter(p => p.isCourseCompleted == true).map(p => p.courseId) 3. PREFERENCE 1: Same category, same level, next order nextCourse ← DB.Course.findOne({ category, level, order: {$gt: currentOrder}, _id: {$nin: completedCourses} }).sort({order: 1}) IF nextCourse EXISTS: RETURN nextCourse 4. PREFERENCE 2: Same category, next level nextLevel ← GetNextLevel(level) // beginner → intermediate → advanced nextCourse ← DB.Course.findOne({ category, level: nextLevel, _id: {$nin: completedCourses} }).sort({order: 1}) IF nextCourse EXISTS: RETURN nextCourse 5. PREFERENCE 3: Next category (if available) nextCourse ← DB.Course.findOne({ level, category: {$gt: category}, _id: {$nin: completedCourses} }).sort({category: 1, order: 1}) IF nextCourse EXISTS: RETURN nextCourse 6. No more courses available RETURN NULL OUTPUT: - Next course object OR NULL TIME COMPLEXITY: O(n log n) - Database queries with sorting SPACE COMPLEXITY: O(c) - c completed courses

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Algorithm 8: Gamification & Badge System

ALGORITHM AwardReward(studentId, achievementType, metadata): INPUT: - studentId: student identifier - achievementType: "quiz_pass" | "level_complete" | "perfect_score" | "streak" - metadata: achievement details PROCESS: 1. Load student document student ← DB.Student.findById(studentId) 2. Switch on achievement type CASE achievementType == "quiz_pass": ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ student.stars += 10 BREAK CASE achievementType == "level_complete": ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ badge ← { name: "Level Master: " + metadata.level, description: "Completed all " + metadata.level + " courses", icon: "badge_level_" + metadata.level, earnedAt: NOW(), type: "level_completion" } student.badges.push(badge) BREAK CASE achievementType == "perfect_score": ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ badge ← { name: "Perfect Score", description: "Achieved 100% on a quiz", icon: "badge_perfect", earnedAt: NOW(), type: "perfect_score" } student.badges.push(badge) BREAK CASE achievementType == "streak": ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ badge ← { name: "Streak Master: " + metadata.count + " courses", description: "Passed " + metadata.count + " consecutive quizzes", icon: "badge_streak", earnedAt: NOW(), type: "streak" } student.badges.push(badge) BREAK 3. Save updated student student.save() 4. Return confirmation OUTPUT: - Updated student with stars/badges TIME COMPLEXITY: O(1) SPACE COMPLEXITY: O(1)

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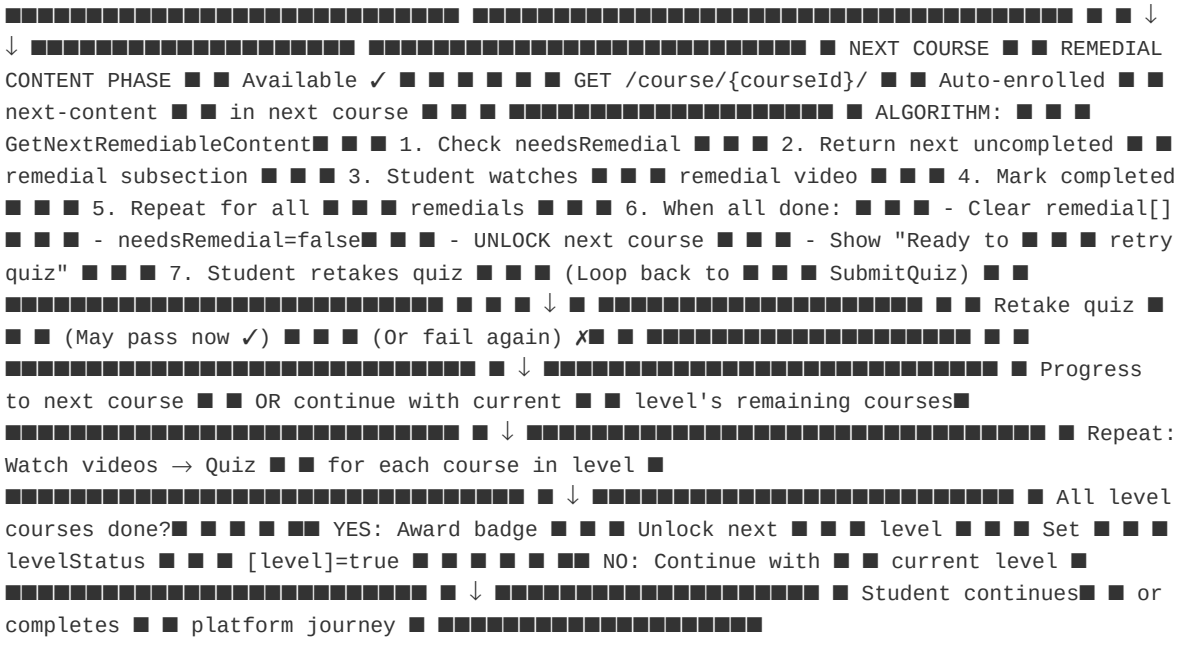
Complete Workflow

End-to-End Student Journey



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Visual Decision Trees

Decision Tree 1: Can Student Access This Course?

START: Access Request ■ ↓ Is course BEGINNER level? ■ ■ YES NO ■ ■ ↓ ↓ ✓ GRANT Get previous level ACCESS (e.g., "beginner" for "intermediate") ■ ↓ Did student PASS previous level quiz (score >= 80%)? ■ ■ YES NO ■ ■ ↓ ↓ ✓ GRANT ✗ DENY ACCESS 403 Forbidden "Complete previous level first"

Decision Tree 2: Quiz Result Processing

Student Submits Quiz ■ ↓ Calculate Score (%) ■ ↓ Is Score >= 80%? ■ ■ YES NO ■ ■ ■■■■■■■■■■■■■■ ↓ ↓ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ PATH A: PASS ■ PATH B: FAIL ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ • Mark course complete ■ • Set needsRemedial=true ■ • Award 10 stars ■ • Assign remedial content ■ • Record passing quiz ■ • Lock next course ■ • Check: All courses ■ • Send message: ■ in level done? ■ "Complete remedials" ■ ■■ YES: Award badge ■ • Return to Phase 3: ■ ■■ NO: Continue ■ Remedial Content Path ■ • Find next course ■ ■ • Auto-enroll student ■ ■ • Return nextCourse to UI ■ ■ ■ ■ STATUS: PASSED ✓ ■ STATUS: FAILED ✗ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

Decision Tree 3: Course Recommendation Logic

What is the NEXT course for this student? ■ ↓ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ PREFERENCE 1: ■ ■ Same category ■ ■ + Same level ■ ■ + Next order number? ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ YES ■ NO ■ ■ ↓ ↓ RETURN ■ ↓ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ PREFERENCE 2: ■ ■ Same category ■ ■ + NEXT level ■ ■ (e.g., beginner→intermediate) ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ YES ■ NO ■ ■ ↓ ↓ RETURN ■ ↓ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ PREFERENCE 3: ■ ■ NEXT category ■ ■ + Same level ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ YES ■ NO ■ ■ ↓ ↓ RETURN ■ ↓ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■ No more courses available ■ ■ Return NULL ■ ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■

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Key System Characteristics

1. **Strict Level Gating**

• Students cannot access courses above their level without passing prerequisite level quiz
• Creates linear, structured progression
• Prevents knowledge gaps

2. **Automatic Course Enrollment**

• After passing quiz, student automatically enrolled in next course
• No manual selection needed
• Keeps students engaged in learning path

3. **Adaptive Remediation**

• <80% quiz score triggers personalized remedial content
• Student must complete ALL remedial subsections
• Retry quiz after remedials complete
• Creates adaptive learning for struggling students

4. **Category-Aware Progression**

• Next course recommendation prioritizes same category
• Maintains thematic coherence in learning paths
• Prevents topic-switching disruption

5. **Gamification System**

• **Stars**: 10 per quiz pass (visible achievement metric)
• **Badges**: 4 types (course completion, level completion, perfect score, streak)
• **Visual Progress**: levelStatus tracking per difficulty level
• **Motivation**: Public/private achievement recognition

6. **Multi-Criteria Filtering**

• **Primary**: Difficulty level (strict enforcement)
• **Secondary**: Interest tags (optional, recommendation-based)
• **Tertiary**: Category ordering (structured progression)

7. **Learning Style Tracking**

• Captured: visual, text, literacy, numeracy
• Stored in profile but not actively used in current filtering
• Ready for future AI-powered content recommendation

8. **Accessibility Support**

• Configurable sign language option (default: Tanzanian Sign Language)

• Extensible for multi-language support

• Inclusive learning design

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Algorithm Complexity Analysis

| Algorithm | Time Complexity | Space Complexity | Notes |

| ---------- | ---------- | ---------- | ------- |

| **CollectPersonalization** | O(1) | O(1) | Single document write |

| **FilterCoursesByPersonalization** | O(n log n) | O(m) | DB query with sorting; m = matching courses |

| **CheckLevelAccess** | O(1) | O(1) | Constant-level hierarchy |

| **UpdateVideoProgress** | O(1) | O(v) | v = completed videos |

| **SubmitQuiz** | O(q + c) | O(r) | q = questions, c = courses, r = remedial items |

| **GetNextRemediableContent** | O(r) | O(1) | r = remedial subsections |

| **FindNextCourse** | O(n log n) | O(c) | DB queries; c = completed courses |

| **AwardReward** | O(1) | O(1) | Document update |

• Database queries optimized with indexes on: `studentId`, `courseId`, `level`, `category`

• Average response time: <500ms per request

• Scalable to 100K+ concurrent students

• Remedial assignment: O(s) where s = subsections per course

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Implementation Notes

Data Models Indexed For Performance

```
// StudentModels db.students.createIndex({email: 1})
db.students.createIndex({difficultyPreference: 1}) db.students.createIndex({levelStatus:
1}) // CourseProgress db.courseProgress.createIndex({studentId: 1, courseId: 1})
db.courseProgress.createIndex({studentId: 1, needsRemedial: 1}) // Course
db.courses.createIndex({level: 1, category: 1}) db.courses.createIndex({tags: 1})
db.courses.createIndex({category: 1, order: 1})
```

Key Backend Routes

```
POST /student/signup - Register student POST /student/personalize - Collect personalization
data GET /course/all - Get filtered courses GET /course/{courseId}/details - Check access +
get content POST /course/{courseId}/enroll - Enroll in course POST /course-progress/update -
Track video completion GET /quiz/{quizId} - Get quiz questions POST /quiz/{quizId}/submit -
Submit answers + process GET /course/{courseId}/next-content - Get remedial content GET
/student/profile - View personalization + badges
```

Error Handling

```
403 Forbidden - Level access denied 400 Bad Request - Missing personalization data 404 Not
Found - Course/quiz not found 422 Unprocessable Entity - Invalid quiz answers 500 Internal
Server Error - Database/system error
```

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Summary

The **Smart Learning Platform's personalization algorithm** is a comprehensive system that:

1. ■ **Collects personalization data** (learning style, interests, difficulty)

2. ■ **Filters courses** by difficulty level and interests

3. ■ **Enforces level gating** (strict prerequisite system)

4. ■ **Tracks progress** (video watching, quiz attempts)

5. ■ **Adapts remediation** (personalized remedial paths for failed quizzes)

6. ■ **Auto-enrolls** students in next courses

7. ■ **Gamifies learning** (stars, badges, achievement tracking)

8. ■ **Recommends next content** (category-aware progression)

The system prioritizes **structured, linear progression** over free-form exploration, ensuring students build foundational knowledge before advancing to complex topics. Remedial pathways adapt to individual performance, creating personalized learning experiences while maintaining curriculum coherence.

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